Abstract workflow Node	Input	Output	Implementation
Reading in both datasets	csv files	csv files	import pandas as pd import numpy as np pd.read_csv(‘ .csv’, sep= ‘,’, encoding = ‘utf-8’) https://pandas.pydata.org/
Treating missing values	csv files	datasets (aus -999 -> NaN)	own implementation
Ensure regional allocation	datasets (without missing values)	cleaned datasets (without lines with missing location assignment)	is.null()
Merge weather and election turnout data by location	cleaned datasets	merged dataframes (with location, temperature, rain, sun, election turnout)	pd.merge(weather_df, election_df, on="location") https://pandas.pydata.org/
test correlation	merged dataframes (with location, temperature, rain, sun, election turnout)	correlation coefficients, p-value	.corr() 3x for each location: election turnout - temperature, election turnout - rain, election turnout - sun
Compare Groups (temperature, rain, sun)	correlation coefficients, p-value	results	pd.groupby() https://pandas.pydata.org/
Results in text format	dataframes	results in text format	own implementation
Visualization of results	dataframes	plots in png format	import matplotlib https://matplotlib.org/ plt.plot(), plt.boxplot(), plt.title(), plt.xlabel(), plt.ylabel()